

PROJECT DOCUMENTATION

A Data-Driven Analysis of Team Performance in Premier League and La Liga (2018-2023)

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Course: Software Project

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1. Overview

This document provides a comprehensive technical and analytical description of the Software Project developed by Shayan Rahmati. The project performs a data-driven comparison of team performance between Premier League and La Liga over a period of five seasons (2018-2023).

Primary objectives:

- Compare attacking output (Goals & xG) across both leagues season by season
- Measure finishing efficiency defined as Goals minus Expected Goals (xG)
- Examine home vs. away performance differences for each team
- Identify seasonal trends

2. Data & Methodology

2.1 Data Sources

Match-level data was collected from two industry-standard football statistics providers:

- FBref (<https://fbref.com>), Goals and Expected Goals (xG) per match
- TransferMarkt (<https://www.transfermarkt.com>), Transfer expenditure by league

The final dataset contains 7600 rows: 3800 for Premier League and 3800 for La Liga.

2.1 Data Preprocessing

All preprocessing was performed in Python. The key steps were:

- Extraction of home/away goals and xG for every match
- Exclusion of the incomplete 2023-2024 season to ensure consistency
- Season labelling: matches from August-December assigned to the current year (e.g., Aug 2019 → 2019-2020); matches from January-May assigned to the prior season start (e.g., Apr 2019 → 2018-2019)
- Calculation of Efficiency: Goals Scored - Expected Goals (xG) per team per match

2.3 Visualization

Processed data was visualized using Tableau, producing bar charts, scatter plots, and efficiency trend charts for comparative analysis.

3. Key Metrics Defined

- **Goals:** Actual goals scored per match (home and away)
- **xG (Expected Goals):** Statistical measure of shot quality, probability each shot results in a goal
- **Efficiency:** Goals - xG; positive = overperformance (clinical finishing), negative = underperformance (home and away)

4. Implementation

4.1 Loading & Parsing Dates

The raw dataset was loaded from a CSV file and the Date column was converted to a proper datetime format to enable season-based filtering.

```
df = pd.read_csv("project2.csv")  
df["Date"] = pd.to_datetime(df["Date"])
```

4.2 Season Labelling

A custom function was applied to assign each match to the correct season. Matches from August onwards belong to the season starting that year and matches before August belong to the season that started the previous year.

```
def get_season(date):  
    if date.month >= 8:  
        return str(date.year) + '-' + str(date.year + 1)  
    else:  
        return str(date.year - 1) + '-' + str(date.year)  
df['Season'] = df['Date'].apply(get_season)
```

4.3 Extracting Home & Away Data

Goals were extracted from the Score column by splitting on the dash separator. Home and away records were then separated into two DataFrames, each with a Venue label.

```
df[['home_goals', 'away_goals']] = df['Score'].str.split('-',  
    expand=True).astype(float)
```

```
home_df = df[["Season", "Home", "home_goals", "xG"]].copy()  
home_df.columns = ["Season", "Team", "Goals", "xG"]  
home_df["Venue"] = "Home"
```

```
away_df = df[["Season", "Away", "away_goals",  
    "xG.1"]].copy()  
away_df.columns = ["Season", "Team", "Goals", "xG"]  
away_df["Venue"] = "Away"
```

4.4 Calculating Efficiency

Finishing efficiency was computed as Goals minus Expected Goals (xG) for both home and away records, then merged into a single dataset.

```
home_df["Efficiency"] = home_df["Goals"] - home_df["xG"]  
away_df["Efficiency"] = away_df["Goals"] - away_df["xG"]  
teams_df = pd.concat([home_df, away_df])
```

5. Key Findings

- The Premier League showed higher offensive intensity than La Liga, with goals exceeding xG in multiple seasons, partly explained by significantly higher transfer investment (€2.2B vs €1.0B average per season).
- xG proved a reliable predictor of attacking output in both leagues, with a strong positive correlation between chance creation and goals scored across all five seasons.
- Premier League teams demonstrated greater adaptability in away matches, while La Liga teams relied more heavily on home advantage for finishing efficiency.

- Strong finishing performance alone does not guarantee league success, Barcelona recorded their highest efficiency in 2019-2020 without winning La Liga, and won the title in 2022-2023 with negative efficiency.
- Managerial and squad continuity were key drivers of efficiency: Liverpool and Atlético Madrid stayed consistently efficient throughout the period, while Chelsea declined sharply after a full managerial overhaul and Barcelona dropped after Messi's departure.

6. Limitations

While this study provides meaningful insights into team performance, several limitations should be acknowledged:

- Offensive metrics only: The dataset is limited to attacking statistics (Goals and xG). Defensive metrics such as Expected Goals Against (xGA), tackles, and clean sheets are not included, restricting a full assessment of overall team quality.
- No possession or pressing data: Metrics like possession percentage or PPDA (Passes Allowed Per Defensive Action) were not incorporated, which may explain some efficiency differences between teams.
- Average league transfer expenditure (€2.2B vs €1.0B) was used as an indicator of player quality, but does not account for squad depth or player age.
- The effect of promoted and relegated teams are not taken into account.